

## Experiment No: 6

**Title:** Use string functions and regex for email validation, data extraction, and text analysis.

**Software/Tools Required:** JavaScript, HTML, CSS and Code Editor.

### Theory:

#### String Functions and Regular Expressions (Regex) in JavaScript

Strings are one of the most commonly used data types in JavaScript and are used to store and manipulate textual data. JavaScript provides several built-in string functions that make it easy to perform operations such as searching, extracting, replacing, and analyzing text. Regular Expressions (Regex) are powerful patterns used to match, validate, and extract specific information from strings. Together, string functions and regex play an important role in data validation, text processing, and information extraction.

#### 1. String Functions in JavaScript

A string is a sequence of characters enclosed within single quotes ("), double quotes (""), or backticks (`).

- `length` – Returns the number of characters in a string.
- `toUpperCase()` and `toLowerCase()` – Convert a string to uppercase or lowercase.
- `substring()` – Extracts a portion of a string.
- `indexOf()` – Returns the position of a specified character or word.
- `replace()` – Replaces specified text with another value.
- `split()` – Converts a string into an array based on a separator.

#### 2. Regular Expressions (Regex)

A Regular Expression is a sequence of characters that defines a search pattern. Regex is commonly used for validation, searching, matching, and extracting text from strings.

- `test()` – Checks whether a pattern exists in a string.
- `match()` – Returns matched values.
- `search()` – Returns the position of a match.
- `replace()` – Replaces matched text.

## **Case Study:**

**Title:** AI Student Profile Parser

**Problem Statement:** Develop a simple JavaScript-based text processing application that accepts an unstructured student introduction and extracts useful information such as the student's name, roll number, email address, phone number, and department using string functions and Regular Expressions.

**Objective:** To demonstrate the practical use of string functions and Regex for information extraction, validation, and basic text analysis.

## **Working:**

- The user enters a paragraph containing student information.
- Regex is used to identify the student's name, roll number, email address, phone number, and department.
- The extracted email address and phone number are checked for valid formats.
- String functions are used to calculate the number of words and characters.
- The entered text can be converted into lowercase for normalization.
- The department name can be transformed into an alternative label.
- The extracted information and analysis results are displayed in a structured format.

**Result:** The student profile information is successfully extracted from unstructured text. Regex performs pattern matching and validation, while JavaScript string functions perform text transformation and analysis.

CASE STUDY • STRING OPERATIONS



## Smart String Studio

Reverse text or perform quick vowel analysis with JavaScript.

Enter String / Paragraph

Arpita Studies at SIT

Select Operation

Count Vowels

Run Operation +

Vowels Found

A • i • a • u • i • e • a • I

Total Vowels: 8

CASE STUDY • STRING OPERATIONS



## Smart String Studio

Reverse text or perform quick vowel analysis with JavaScript.

Enter String / Paragraph

Arpita Studies at SIT

Select Operation

Reverse String

Run Operation +

Original

Arpita Studies at SIT

Reversed

TIS ta seidutS atiprA

## CASE STUDY 02 - REGEX EXTRACTION

# AI Student Profile Parser

Convert an unstructured paragraph into useful student data.

Hi, I am Arpita Waghmare , I study computer science and engineering at Symbiosis Institute of Technology

Extract & Analyze ✨

STUDENT NAME

Arpita Waghmare

ROLL NUMBER

Not Found

EMAIL

Not Found

EMAIL STATUS

✗ Invalid

PHONE

Not Found

PHONE STATUS

✗ Invalid

DEPARTMENT

computer science

TOTAL WORDS

16

CHARACTERS

88

### Lowercase Transformation

hi, i am arpita waghmare , i study computer science and engineering at symbiosis institute of technology

### Department Transformation

Hi, I am Arpita Waghmare , I study Artificial Intelligence & Data Science and engineering at Symbiosis Institute of Technology

PRACTICAL 06 • TEXT PROCESSING

# AI Text Lab ✦

Regex, strings and smart text analysis in one place.

## Input Text

Hello Ai! my mail is arpitawaghmare79@gmail.com and my score is 95%

## Analysis Mode

Regex Data Extraction



Run Text Analysis →

## Regex Extraction

Emails: arpitawaghmare79@gmail.com

Numbers: 95

### Analysis Mode

Find Vowels



Run Text Analysis →

**Vowels:** e • o • A • i • a • i • i • a • i • a • a • a • e • a • i • o • a • o • e • i

Total: 20

### Analysis Mode

Text Statistics



Run Text Analysis →

#### Text Statistics

Characters: 67

Words: 13

Vowels: 20

Digits: 4

### Analysis Mode

Reverse Text



Run Text Analysis →

#### Reversed:

%59 si erocs ym dna moc.liamg@97eramhgawatipra si liam ym !iA olleH

**Analysis Mode**

Palindrome Check

Run Text Analysis →

✕ No — not a palindrome

## Conclusion:

String functions and Regular Expressions are essential tools for text processing in JavaScript. String methods provide efficient ways to manipulate and analyze textual data, while regex enables pattern matching, validation, and data extraction. By combining these techniques, applications such as email validation, data extraction systems, and text analysis tools can be developed effectively and accurately.